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Optimization of entropy generation in Carreau–Yasuda hybrid nanofluid flow: the combined influence of non-Fourier heat flux, couple stress, and Soret–Dufour

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ABSTRACT

In this study, we examined the steady movement of Carreau-Yasuda hybrid nanofluid across an irregularly extended sheet influenced by the magnetic field, non-Fourier heat flux, couple stress effects, and the synergistic impacts of Soret and Dufour phenomena. The governing partial differential equations are transformed into a system of nonlinear ordinary differential equations using appropriate similarity variables and then solved using the bvp4c solver. It is detected that the Nusselt number drops by 13.30% when the Dufour number falls between 0 to 1 and the rate of mass transmission drops by 9.69% when the Soret number falls between 0 to 1.

KEYWORDS

Non-Newtonian fluid; bvp4c; multiple linear regression; thermal radiation; Biot number

1. Introduction

Non-Newtonian hybrid nanofluids hold immense significance due to their enhanced thermal properties compared to conventional fluids. These fluids, composed of a base fluid, multiple types of nanoparticles, and exhibiting non-Newtonian behavior, find applications in various fields. In heat transfer processes, they improve efficiency and reduce energy consumption. In biomedical engineering, they aid in targeted drug delivery and thermal therapy. Additionally, their superior lubrication properties make them suitable for enhancing the performance of mechanical systems. Along the lubricated surface with viscous dissipation, Ahmad and Nadeem [1] assessed the radiative hybrid nanofluid using the Casson fluid framework. Nazir *et al.* [2] examined the non-Darcy movement of the Carreau–Yasuda (CY) hybrid nanofluid incorporating ion slip and heat source characteristics by an identical numerical approach. An upsurge in the Weissenberg number correlates with an escalation in the friction factor. Arif *et al.* [3] analyzed the movement of a Maxwell hybrid nanofluid through a cylinder that oscillates vertically. Wang *et al.* [4] inspected the dynamics of CY ternary hybrid nanofluid on an upward-facing surface considering magnetic dipole and thermal source parameters. It is noted that the magnetic dipole is significantly involved in the process of increasing the fluid's temperature. Afterwards, various authors Alqahtani *et al.* [5] have analyzed energy transmission through Carreau Yasuda fluid influenced by ethylene glycol with activation energy and ternary hybrid nanocomposites by using a mathematical

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model. Algehyne *et al.* [6] have reviewed analysis of thermally stratified micropolar CY hybrid nanofluid flow with Cattaneo–Christov heat and mass flux. Jubair *et al.* [7] have explored investigating thermal conduction dynamics in ternary CY nanofluids under convective boundary conditions and exponential heat source/sink effects. Hayat *et al.* [8] have possessed a thermal conductivity analysis for peristalsis of hybrid nanofluid with Darcy–Forchheimer law. Murugan *et al.* [9] have explored entropy generation on MHD motion of hybrid nanofluid with porous medium in presence of thermo-radiation and ohmic viscous dissipation. Ahmed *et al.* [10] have possessed thermal performance of radiative hydromagnetic peristaltic pumping of hybrid nanofluid through curved duct with industrial applications. Gomathi and Poulomi [11] have discussed entropy optimization on EMHD Casson Williamson penta-hybrid nanofluid over porous exponentially vertical cone. De [12] has studied entropy optimization of stagnant blood flow systems with tetra-hybrid nano additives under viscous dissipation, joule heating, and thermal radiation effects.

The Cattaneo (C)–Christov (C) heat flow model, an extension of Fourier's law, incorporates a relaxation time parameter, accounting for the finite propagation speed of thermal signals. This model is crucial for studying heat transfer in materials with high thermal conductivity or in situations where rapid temperature changes occur. Applications include nanofluid flows, microfluidics, and materials processing, where accurate prediction of heat transfer is essential for optimizing performance and controlling thermal stresses. Using a non-Fourier heat flux model, Rawat and Kumar [13] scrutinized the movement of aqueous nanofluid across a thermally radiated elastic surface. Compared to the non-Fourier's approach, Fourier's approach shows that temperature has a more significant role. Ali *et al.* [14] explored the Galerkin method-based rotating chemically reactive movement of Maxwell nanofluid through an elastic sheet using the same model. A drop in fluid temperature is observed when the thermal relaxation parameter surges. Using the Keller–Box approach, Abu-Hamdeh *et al.* [15] inspected the movement of nanofluid (engine oil based) across an elastic surface in the presence of a heat source and a non-Fourier heat flux. John Christopher *et al.* [16] used the RKF-45 procedure to study the non-Darcy movement of a hybrid nanofluid in an elastic cylinder with non-Fourier heat flux model. It has been revealed that when the Forchheimer parameter is amplified, there is a decline in the velocity of the fluid. The non-Newtonian Powell–Eyring binary/mono nanofluid transport over a porous medium was inspected by Waqas *et al.* [17]. A decline in fluid velocity is observed with an upsurge in the porosity parameter. The motion of a non-Fourier heat flux-driven convectively heated upright cone through which an aqueous Maxwell (non-Newtonian) hybrid nanofluid passed was studied by Chandra Sekar Reddy *et al.* [18]. It has been observed that when the Deborah number surges, so does the heat transfer rate. Subsequently, the non-Fourier heat flux model was employed by several authors [19–22] to examine different fluid dynamics across diverse surfaces.

Couple stress fluids, which exhibit size-dependent behavior, are characterized by their ability to resist rotational deformation. The concept of couple stress, which accounts for the rotational motion of fluid particles, is significant in understanding the behavior of fluids with microstructure, such as polymeric solutions, colloidal suspensions, and biological fluids. Applications of couple stress fluid flows include microfluidics, lubrication theory, and biofluid mechanics. By incorporating couple stress effects into the governing equations, researchers can better model and predict the behavior of these complex fluids in various engineering and biological systems. Mahesh *et al.* [23] explored the impact of the couple stress parameter on the movement of radiative hybrid nanofluid through a permeable shrinking surface. A fall in fluid temperature is noted with a surge in the couple stress factor. Later, the couple stress model was employed by several authors [24–26] to scrutinize various fluid dynamics across different surfaces.

The Soret and Dufour effects are particularly substantial in systems with large density variations, such as those involving multicomponent mixtures or non-ideal fluids. The Soret effect has applications in isotope separation, chemical separation processes, and the study of atmospheric and oceanic flows. The Dufour effect, while less commonly studied, is relevant in energy transport processes in materials with significant thermal and concentration gradients, such as in certain types of heat exchangers and energy storage systems. Ibrahim and Zemedu [27] studied the movement of a micropolar fluid along an elastic surface utilizing the bvp4c solver. It is detected that when the Dufour number surges, the fluid's temperature escalates. Singha *et al.* [28] analyzed the movement of aqueous nanofluid across an elastic surface incorporating cross-diffusion effects by the shot method. It has been noticed that there is a correlation between the surge in the Soret number and the rise in the concentration. Jawad *et al.* [29] used HAM to examine the formation of entropy in the movement of a Maxwell nanofluid across an elastic sheet. It is detected that there an escalation in the permeability parameter is associated with a decline in entropy generation. Subsequently, further studies [30–34] examined diverse fluid dynamics on various surfaces demonstrating cross-diffusion effects.

Thermal radiation considerably influences heat transfer rates and temperature distributions in fluid flows, particularly at elevated temperatures. This is essential in applications such as nuclear reactor cooling, solar energy systems, spacecraft thermal control, and furnace design. Nisar *et al.* [35] scrutinized the magnetohydrodynamic peristaltic flow of a Bingham nanofluid within a symmetric channel, taking into account thermal radiation and convective boundary conditions. The study demonstrated that the fluid's temperature diminishes with an increase in the thermal radiation parameter, underscoring the substantial importance of radiation as a cooling process. Wang *et al.* [36] analyzed the entropy generation in magnetohydrodynamic Casson nanofluid flow comprising carbon nanotubes, copper oxide, and titanium dioxide nanoparticles suspended in a sodium alginate base fluid across a stretching sheet. The research indicated that elevating the thermal radiation parameter substantially enhances the heat transfer rate from the surface. Nisar *et al.* [37] investigated the magnetohydrodynamic peristaltic flow of a Bingham nanofluid within a symmetric, compliant-walled tube, including the effects of thermal radiation. Later, distinct authors [38–42] examined the impact of thermal radiation on various fluid flows.

While extensive research has been conducted on the flow and heat transfer of hybrid nanofluids, studies considering the combined influence of non-Fourier heat flux, couple stress effects, and Soret–Dufour phenomena within the framework of CY fluid flow over a nonlinear stretching sheet are limited. This study seeks to fill that knowledge gap by thoroughly examining the complex relationship between these aspects. Furthermore, this study employs multiple linear regression analysis to quantify the impact of key parameters (such as Dufour number and Soret number) on crucial performance indicators like the skin friction coefficient. The findings of this research have significant implications for various industrial applications. The results provide valuable insights for optimizing polymer extrusion processes, enhancing heat transfer in metal rolling operations, and improving the performance of microfluidic devices used in drug delivery and lab-on-a-chip technologies. Furthermore, the study contributes to a better understanding of heat transfer phenomena in systems involving non-Newtonian fluids, which is crucial for the conceptualization and creation of efficient heat exchangers and cooling systems in various industries, including electronics, automotive, and aerospace.

1.1. Research questions

To ensure the study remains focused on its novel contributions, the following research questions are formulated to directly address the gaps and unique aspects of the investigation:

- What is the impact of the couple stress parameter and volume fraction of nanoparticles on the velocity profile?
- What is the impact of the Dufour number, thermal radiation parameter, and thermal relaxation parameter on the temperature profile of the fluid?
- How do the Soret number and chemical reaction parameter affect the concentration profile of the nanofluid?
- What are the effects of key parameters, such as the Dufour number, Brinkman number, and volume fraction of nanoparticles, on the entropy generation in the fluid flow?
- What are the quantitative percentage changes in key engineering design parameters (skin friction coefficient, Nusselt number, and Sherwood number) due to variations in critical factors like the couple stress, Dufour, and Soret numbers?

2. Formulation of the study

This research looks at the steady movement of CY hybrid nanofluid through a nonlinear elastic sheet with non-Fourier heat flux and how several parameters, such as Dufour, couple stress, and Soret affect it. In this investigation, we presuppose the following:

i. Please refer to Table 1 for the precise estimates of the thermo-physical features of $WEG(H_2O + EG)$, TiO_2, and Cu.
ii. A vertically imposed external magnetic field of strength B affects the flow (see Fig. 1).
iii. Viscous dissipation and induced magnetic field parameters are ignored in this study.
iv. The gravitational force is often neglected due to its relatively small magnitude compared to other dominant forces like inertial and viscous forces.

The subsequent circumstances and equations are requisite for this study, predicated on the aforementioned assumptions [44, 45, 46, 47]:

$$\frac{\partial v}{\partial y} + \frac{\partial u}{\partial x} = 0 \tag{1}$$

Table 1. Key parameters' values of $WEG(H_2O + EG)$, TiO_2, and Cu [43]

Properties	$WEG(H_2O + EG)(f)$	$Cu\ (\phi_C)$	$TiO_2\ (\phi_T)$
$\sigma(Sm^{-1})$	2.06×10^{-6}	59.6×10^{6}	2.6×10^{6}
$k(Wm^{-1}K^{-1})$	0.1816	401	8.9538
$\rho(kgm^{-3})$	1041.89	8933	4250
$C_p(Jkg^{-1}K^{-1})$	3421.54	85	686.2

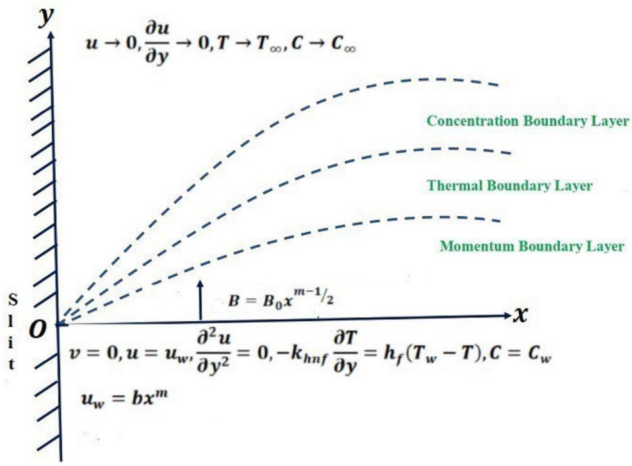

Figure 1. Diagrammatical representation of this study.

$$\frac{\partial u}{\partial x}u + \frac{\partial u}{\partial y}v = \frac{\partial^2 u}{\partial y^2}\upsilon_{hnf} + \upsilon_{hnf}\Gamma^d\left(\frac{n-1}{d}\right)(d+1)\left(\frac{\partial u}{\partial y}\right)^d\frac{\partial^2 u}{\partial y^2} - \left(\sigma_{hnf}uB^2 + \frac{\partial^4 u}{\partial y^4}\gamma_0\right)\frac{1}{\rho_{hnf}} \tag{2}$$

$$v\frac{\partial T}{\partial y} + u\frac{\partial T}{\partial x} = \left(\left(k_{hnf} + \frac{16}{3}\frac{\sigma * T_\infty^3}{k*}\right)\frac{\partial^2 T}{\partial y^2}\right)\frac{1}{(\rho C_p)_{hnf}} - \delta\begin{bmatrix} u^2\frac{\partial^2 T}{\partial x^2} + 2uv\frac{\partial^2 T}{\partial x \partial y} + v^2\frac{\partial^2 T}{\partial y^2} + \\ u\frac{\partial T}{\partial x}\frac{\partial u}{\partial x} + v\frac{\partial T}{\partial x}\frac{\partial u}{\partial y} + u\frac{\partial T}{\partial y}\frac{\partial v}{\partial x} + v\frac{\partial T}{\partial y}\frac{\partial v}{\partial y}\end{bmatrix} + \frac{D_m k_T}{c_s C_p}\frac{\partial^2 C}{\partial y^2} \tag{3}$$

$$v\frac{\partial C}{\partial y} + u\frac{\partial C}{\partial x} = \frac{\partial^2 C}{\partial y^2}D_m - k_0(C - C_\infty) + \frac{D_m k_T}{T_m}\frac{\partial^2 T}{\partial y^2} \tag{4}$$

$$\left.\begin{aligned} &\text{at } y = 0 : C = C_w \ , u = u_w = bx^m, \frac{\partial^2 u}{\partial y^2} = 0, v = 0, \frac{\partial T}{\partial y}k_{hnf} = (T - T_w)h_f, \\ &\text{as } y \to \infty : T \to T_\infty, \frac{\partial u}{\partial y} \to 0, C \to C_\infty, u \to 0. \end{aligned}\right\} \tag{5}$$

For the purpose of converting governing equations, Algehyne *et al.* [48] introduced the following similarity transformations:

$$\left.\begin{aligned} &\eta = yx^{\frac{m-1}{2}}\sqrt{\frac{m+1}{2}}\sqrt{\frac{b}{\upsilon}}, v = -\left[f(\eta) + \eta\left(\frac{m-1}{m+1}\right)f'(\eta)\right]\sqrt{\frac{m+1}{2}}\sqrt{b\upsilon}x^{\frac{m-1}{2}}, \\ &C - C_\infty = \Phi(\eta)(C_w - C_\infty), u = bx^m f'(\eta), \theta(\eta) = \frac{T - T_\infty}{T_w - T_\infty}. \end{aligned}\right\} \tag{6}$$

Equation (1) can be readily satisfied by the terms presented in (6). The successful modification of Equations (2)–(4) and conditions in (5) through integration of (6) yields the following result:

$$\frac{m+1}{2}\frac{1}{S_1S_2}f''' + \frac{d+1}{d}\frac{1}{S_1S_2}\left(\frac{(m+1)(n-1)}{2}\right)We^d(f'')^d f''' \\ -\frac{S_3}{S_1}Mgf' - \left(\frac{m+1}{2}\right)^2\frac{1}{S_1}Csf^v + \frac{m+1}{2}ff'' - mf'2 = 0 \tag{7}$$

$$\frac{m+1}{2}\left(\frac{S_4}{S_5}+\frac{Ra}{S_5}\right)\frac{1}{\Pr}\theta'' - \Upsilon\left(\frac{m+1}{2}\right)\left(\frac{m+1}{2}\right)f^2\theta'' + \frac{1}{2}(m+1)f\theta' \\ +\Upsilon\left(\frac{m^2-1}{4}ff'\theta' + \left(\frac{m-1}{2}\right)\left(\frac{m-1}{2}\right)\eta^2 f'2\theta'2\right) + Du\Phi'' = 0 \tag{8}$$

$$\frac{m+1}{2}\frac{1}{St}\Phi'' + \frac{m+1}{2}f\Phi' - Kr\Phi + \frac{m+1}{2}Sr\theta'' = 0 \tag{9}$$

$$\left.\begin{array}{l} \text{at } \eta = 0 : f(\eta) = 0, \Phi(\eta) = 1, f'(\eta) = 1, \theta'(\eta) = -\dfrac{1}{S_4}(1-\theta(\eta))Bi, f'''(\eta) = 0, \\ \text{as } \eta \to \infty : \Phi(\eta) \to 0, f'(\eta) \to 0, \theta(\eta) \to 0, f''(\eta) \to 0. \end{array}\right\} \tag{10}$$

where

$$\left.\begin{array}{l} \Upsilon = \delta_0 b, \delta_0 = \delta x^{m-1}, Mg = \dfrac{\sigma_f B_0^2}{\rho b}, B = B_0 x^{\frac{m-1}{2}}, We = \Gamma_0\sqrt{\dfrac{b(m+1)u_w^2}{2\upsilon}}, \\ \Gamma_0 = \Gamma x^{\frac{m-1}{2}}, Cs = \dfrac{\gamma_1 b}{\rho\upsilon^2}, \gamma_1 = \gamma_0 x^{m-1}, Kr = \dfrac{k_1}{b}, k_0 = k_1 x^{m-1}, Ra = \dfrac{16}{3}\dfrac{\sigma * T_\infty^3}{kk*}, \\ Du = \dfrac{D_m k_T (C_w - C_\infty)}{\upsilon c_s C_p (T_w - T_\infty)}, \Pr = \dfrac{\mu C_p}{k}, St = \dfrac{\upsilon}{D_m}, Sr = \dfrac{D_m k_T (T_w - T_\infty)}{\upsilon T_m (C_w - C_\infty)}, \\ Bi = \dfrac{a}{k_f\sqrt{\frac{b(m+1)}{2\upsilon}}}, h_f = a x^{\frac{m-1}{2}}. \end{array}\right\}$$

and

$$\left.\begin{array}{l} S_5 = (1-\phi_T)\left[(1-\phi_C) + \phi_C\dfrac{(\rho C_p)_C}{(\rho C_p)_f}\right] + \phi_T\dfrac{(\rho C_p)_T}{(\rho C_p)_f}, \\ S_{31} = \dfrac{\sigma_C + 2\sigma_f - 2\phi_C(\sigma_f - \sigma_C)}{\sigma_C + 2\sigma_f + \phi_C(\sigma_f - \sigma_C)}, S_2 = ((1-\phi_C)(1-\phi_T))^{2.5}, \\ S_1 = \phi_T\dfrac{\rho_T}{\rho_f} + (1-\phi_T)\left[(1-\phi_C) + \phi_C\dfrac{\rho_C}{\rho_f}\right], S_4 = S_{41} \times \dfrac{k_T - 2\phi_T(S_{41}k_f - k_T) + 2S_{41}k_f}{k_T + \phi_T(S_{41}k_f - k_T) + 2S_{41}k_f}, \\ S_3 = S_{31} \times \dfrac{\sigma_T + 2S_{31}\sigma_f - 2\phi_T(S_{31}\sigma_f - \sigma_T)}{\sigma_T + 2S_{31}\sigma_f + \phi_T(S_{31}\sigma_f - \sigma_T)}, \\ S_{41} = \dfrac{k_C - 2\phi_C(k_f - k_C) + 2k_f}{k_C + \phi_C(k_f - k_C) + 2k_f}. \end{array}\right\}$$

The equation for the friction factor is as follows:

$$Cf = \frac{2}{\rho}\left(\frac{\partial u}{\partial y}\right)\frac{\mu_{hnf}}{{u_w}^2}\bigg|_{y=0}. \tag{11}$$

Utilizing (6) allows for a simplified formulation of the formula used in Equation (11) as:

$$\sqrt{\mathrm{Re}_x}Cf = \sqrt{2(m+1)}\frac{1}{S_2}f''(0).$$

To figure out the Nusselt and Sherwood numbers, one can utilize the following approaches:

$$Sh = \frac{-x}{(C_w - C_\infty)}\frac{s_w}{D_m}\bigg|_{y=0}, Nu = \frac{-xq_w}{k_f(T_w - T_\infty)}\bigg|_{y=0}$$
$$\text{where } q_w = \left(k_{hnf} + \frac{4}{3}\frac{4\sigma * T_\infty^3}{k*}\right)\frac{\partial T}{\partial y}, s_w = D_m\frac{\partial C}{\partial y}. \tag{12}$$

By employing (6), the formulas utilized in (12) can be succinctly reformulated as:

$$(\mathrm{Re}_x)^{-1/2}Nu = -\sqrt{\frac{m+1}{2}}(S_4 + Ra)\theta'(0), (\mathrm{Re}_x)^{-1/2}Sh = -\sqrt{\frac{m+1}{2}}\Phi'(0).$$

2.1. Irreversibility formulation

The formula that expresses the dimensional calculation of entropy generation as follows:

$$S_g = \left(\frac{k_{hnf}}{k_f} + 16\frac{\sigma * T_\infty^3}{3k_f k*}\right)\left(\frac{k_f}{T_\infty^2}\right)\left(\frac{\partial T}{\partial y}\right)^2 + \mu_{hnf}\frac{1}{T_\infty}\left(\frac{\partial u}{\partial y}\right)^2 + \sigma_{hnf}\frac{1}{T_\infty}u^2B^2$$
$$+\frac{1}{C_\infty}\left(\frac{\partial C}{\partial y}\right)^2\overline{R}D_m + \frac{1}{T_\infty}\frac{\partial C}{\partial y}\frac{\partial T}{\partial y}\overline{R}D_m. \tag{13}$$

By adopting (6), we may rewrite Equation (13) as follows:

$$N_{EG} = \frac{m+1}{2}(S_4 + Ra)\alpha\theta'^2 + \frac{m+1}{2}\frac{Br}{S_2}f''^2 + S_3Brf'^2Mg + \frac{m+1}{2}\Lambda\frac{\beta}{\alpha}\Phi'^2 + \frac{m+1}{2}\Lambda\theta'\Phi', \tag{14}$$

where

$$\alpha = \frac{T_w - T_\infty}{T_\infty}, Br = \frac{\mu_f {u_w}^2}{(T_w - T_\infty)k_f}, N_{EG} = \frac{\upsilon T_\infty S_g}{bx^{m-1}(T_w - T_\infty)k_f}, \beta = \frac{C_w - C_\infty}{C_\infty},$$
$$\Lambda = \frac{\overline{R}(C_w - C_\infty)D_m}{k_f}.$$

The Bejan number can be estimated by adopting the subsequent procedure:

$$Be = \frac{\textit{Entropy formation due to the transfer of mass and heat}}{\textit{The overall generation of entropy}}.$$

Applying (14) in the following way allows us to revise Be:

$$Be = \frac{\frac{m+1}{2}(S_4 + Ra)\alpha\theta'^2 + \frac{m+1}{2}\Lambda\frac{\beta}{\alpha}\Phi'^2 + \frac{m+1}{2}\Lambda\theta'\Phi'}{\frac{m+1}{2}(S_4 + Ra)\alpha\theta'^2 + \frac{m+1}{2}\frac{Br}{S_2}f''^2 + S_3Brf'^2Mg + \frac{m+1}{2}\Lambda\frac{\beta}{\alpha}\Phi'^2 + \frac{m+1}{2}\Lambda\theta'\Phi'}.$$

3. Numerical procedure

The built-in function bvp4c in MATLAB is utilized to solve Equations (7)–(9) subject to the restrictions (10). This function effectively handles the given conditions (10) to provide accurate solutions. bvp4c implements a finite difference method based on the Lobatto IIIa formula. This method is known for its efficiency in terms of computational cost. Additionally, it's a robust solver, meaning it can handle a wide range of BVPs without encountering stability issues as often compared to some other techniques. Boundary value problems need to be well-posed for any numerical solver to find a unique and stable solution. bvp4c will not be able to determine if a problem is ill-posed (missing information or has multiple solutions), and it might struggle to converge in such cases.

Before coding, it is necessary to consider the following assumptions.

$$f = f_1, f' = f_2, f'' = f_3, f''' = f_4, f^{iv} = f_5, \theta = f_6, \theta' = f_7, \Phi = f_8, \Phi' = f_9..$$

The following system of first-order ODEs can be derived from Equations (7) to (9) and the conditions specified in (10), as follows [49]:

$$\begin{aligned}
&f_1{}' = f_2,\\
&f_2{}' = f_3,\\
&f_3{}' = f_4,\\
&f_4{}' = f_5,\\
&f_5{}' = \left(\tfrac{2}{m+1}\right)^2 \frac{S_1}{Cs}\left[\begin{array}{c} \dfrac{m+1}{2}\dfrac{1}{S_1S_2}f_4 + \dfrac{d+1}{d}\dfrac{1}{S_1S_2}\left(\dfrac{(m+1)(n-1)}{2}\right)We^d(f_3)^d f_4 \\ -\dfrac{S_3}{S_1}Mgf_2 + \dfrac{m+1}{2}f_1f_3 - mf_2{}^2 \end{array}\right]\\
&f_6{}' = f_7,\\
&f_7{}' = \frac{-4S_5\Pr}{(m+1)\big(2(S_4+Ra) - (m+1)S_5\Pr\Gamma f^2\big)}\\
&\qquad\left[\begin{array}{l} \dfrac{1}{2}(m+1)f_1f_7 \\ +\Upsilon\left(\dfrac{m^2-1}{4}f_1f_2f_7 + \left(\dfrac{m-1}{2}\right)\left(\dfrac{m-1}{2}\right)\eta^2 f_2{}^2 f_7{}^2\right) + Duf_9{}' \end{array}\right],\\
&f_8{}' = f_9,\\
&f_9{}' = \frac{2St}{m+1}\left(\frac{m+1}{2}f_1f_9 - Krf_8 + \frac{m+1}{2}Srf_7{}'\right).
\end{aligned} \tag{15}$$

with the constraints

$$\left.\begin{array}{l} fa(1) = 0, fa(8) = 1, fa(2) = 1, fa(7) = -\dfrac{1}{S_4}\big(1 - fa(6)\big)Bi, fa(4) = 0, \\ fb(8) = 0, fb(2) = 0, fb(6) = 0, fb(3) = 0. \end{array}\right\} \tag{16}$$

Once the system is implemented in MATLAB, the code can be executed to generate the graphical outputs. The procedure for applying the bvp4c solver is outlined in Fig. 2.

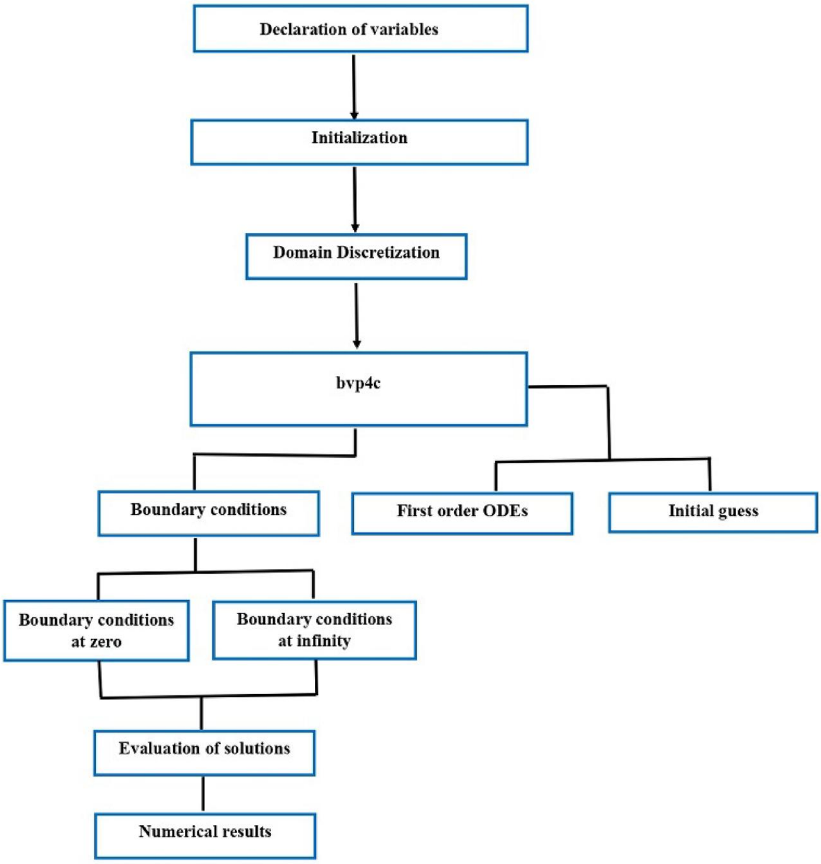

Figure 2. Workflow for implementing the bvp4c algorithm [50].

4. Findings and analysis

4.1. Velocity and other profiles

An upsurge in the Weissenberg number causes a reduction in fluid velocity due to the enhanced shear-thinning behavior of the fluid. With a spike in the parameter, the fluid's apparent viscosity grows, resulting in greater resistance to flow and consequently lower fluid velocity (Fig. 3). An augmentation in the couple stress parameter causes a plunge in the fluid's viscosity, hence diminishing its resistance to flow. This reduced resistance accelerates the fluid's flow, resulting in an upsurge in its velocity (see Fig. 4). This observation has significant practical implications. For instance, in microchannels or narrow gaps where couple stress effects are prominent, adjusting the couple stress parameter can be utilized to control and enhance fluid flow rates. This understanding can aid in the design and optimization of microfluidic devices for various applications, including lab-on-a-chip systems and drug delivery mechanisms. It is apparent from Fig. 5 that there is a drop in the fluid velocity with the escalation in ϕ_C. Nanoparticles act like tiny friction sources within the fluid. As their concentration increases, the fluid experiences more resistance to flow, making it thicker and less likely to move easily. A higher Dufour number indicates a stronger

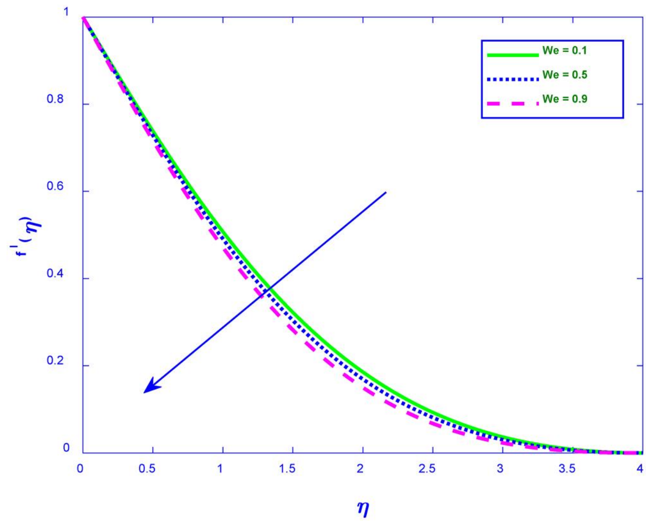

Figure 3. Situation where *We* influences f'. (η)

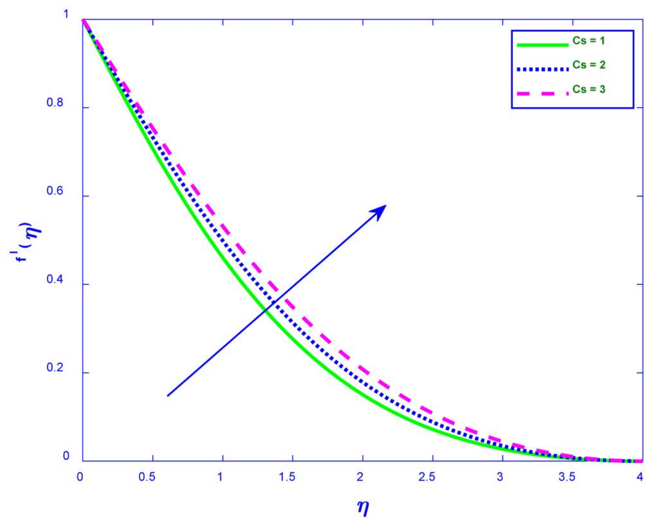

Figure 4. Situation where *Cs* influences f'. (η)

coupling between mass diffusion and heat transfer. As the Dufour number increases, the energy flux due to mass diffusion also increases, causes a more prominent temperature of the fluid (see Fig. 6). This phenomenon finds relevance in various engineering applications, such as in the design of heat exchangers or in processes involving chemical reactions where precise temperature control is crucial. Thermal relaxation time refers to the average time it takes for a fluid particle to transfer its thermal energy to its surroundings. With a longer relaxation time, the particles have more time between collisions. This allows them to distribute their thermal energy more slowly. This causes the fluid's temperature to drop (see Fig. 7). This principle finds applications in various scenarios, such as controlling heat dissipation in electronic devices or managing temperature in industrial processes. The escalation of thermal radiation results in a temperature elevation of a fluid, since thermal

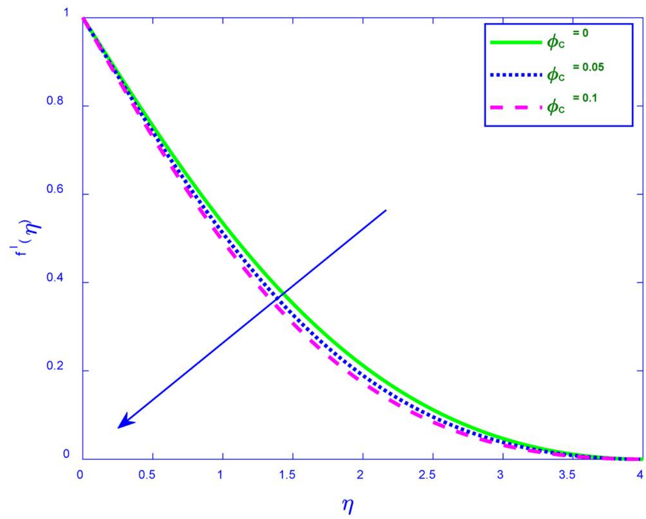

Figure 5. Situation where ϕ_C influences $f'.$ (η)

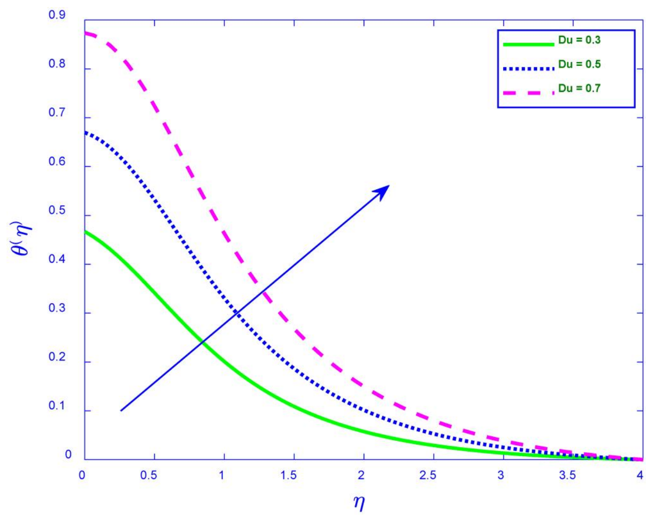

Figure 6. Situation where *Du* influences $\theta.$ (η)

radiation conveys energy *via* electromagnetic waves from a hotter surface or environment to the fluid. When the fluid absorbs this radiation, the energy stimulates its molecules, elevating their kinetic energy. This molecular agitation results in an increase in the temperature of the fluid (see Fig. 8). In practical terms, a rise in the thermal radiation parameter in fluid flow over a nonlinear stretching sheet can lead to several implications. For instance, in industrial processes like film coating or material processing, controlling the thermal radiation can help regulate the temperature of the fluid, influencing the final product's quality. A higher Soret number designates a stronger coupling between temperature and concentration gradients. When the Soret number increases, the temperature gradient becomes more influential in driving mass diffusion. Consequently, the fluid's concentration tends to rise in areas of elevated temperature, resulting in a general augmentation of the concentration

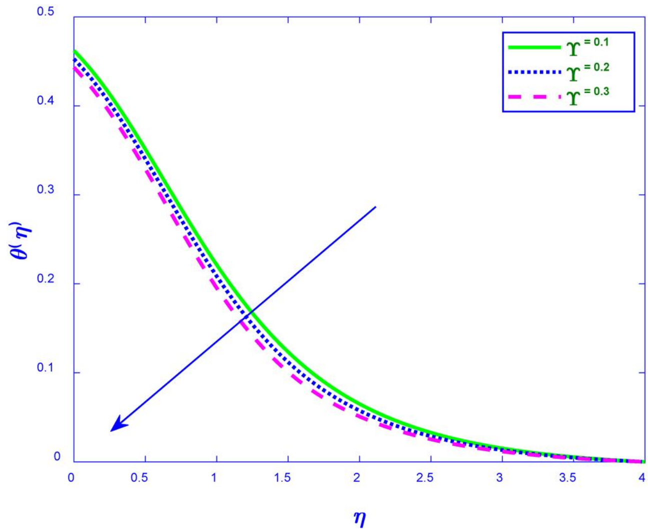

Figure 7. Situation where Y influences $\theta.\ (\eta)$

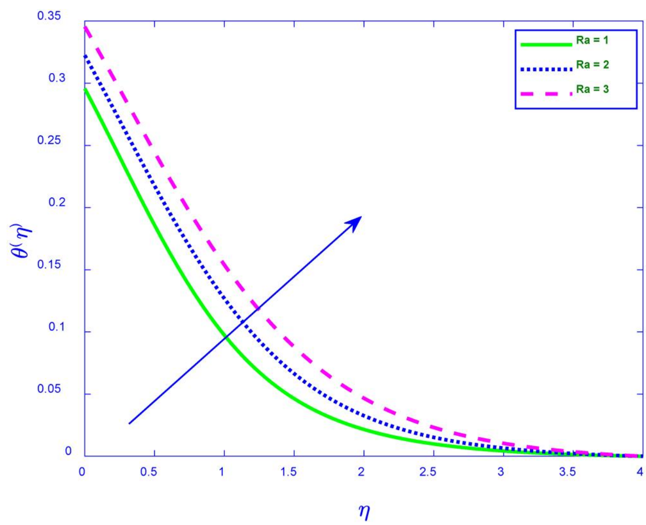

Figure 8. Situation where *Ra* influences $\theta.\ (\eta)$

of the fluid (see Fig. 9). A rise in the chemical reaction parameter can lead to a reduction in fluid concentration due to the increased consumption of reactants within the fluid (see Fig. 10). As the chemical reaction intensifies, more reactants are converted into products, resulting in a decrease in the overall concentration of the original reactants. This phenomenon is particularly relevant in systems where chemical reactions play a significant role in the transport and distribution of substances within a fluid.

4.2. Irreversibility analysis

An upsurge in the Dufour number indicates a stronger coupling between mass and energy transfer, leading to a larger temperature gradient. This enhanced temperature

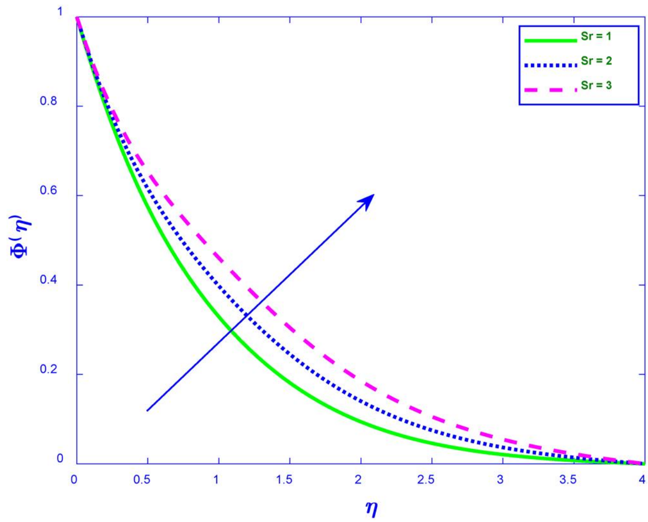

Figure 9. Situation where *Sr* influences $\Phi.\ (\eta)$

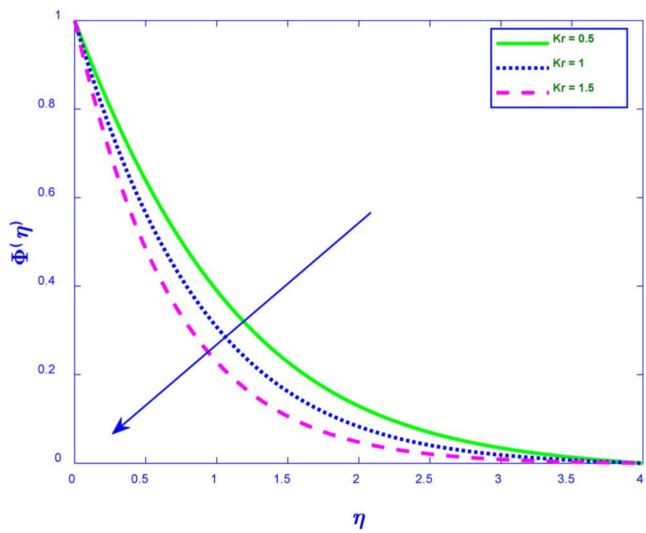

Figure 10. Situation where *K* influences $\Phi.\ (\eta)$

gradient, in turn, drives a greater heat transfer rate, resulting in increased entropy generation within the fluid flow system (see Fig. 11). In practical applications, the relationship between the Dufour number and entropy generation can be utilized in the design and optimization of various engineering systems. For instance, in heat exchangers, understanding how changes in the Dufour number affect entropy generation can aid in improving the efficiency of heat transfer processes. An increase in ϕ_C causes a rise in the entropy generation (see Fig. 12). The thermal conductivity of a nanofluid is proportional to the volume percentage of nanoparticles in it. This leads to a higher rate of heat transfer (when ϕ_C increases), which reduces the heat transfer irreversibility. Consequently, the Bejan number declines, signifying a greater proportion of the entire

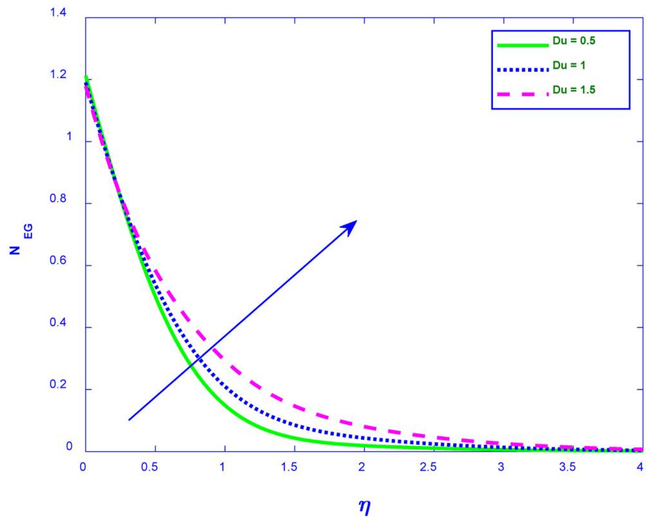

Figure 11. Situation where *Du* influences. N_{EG}

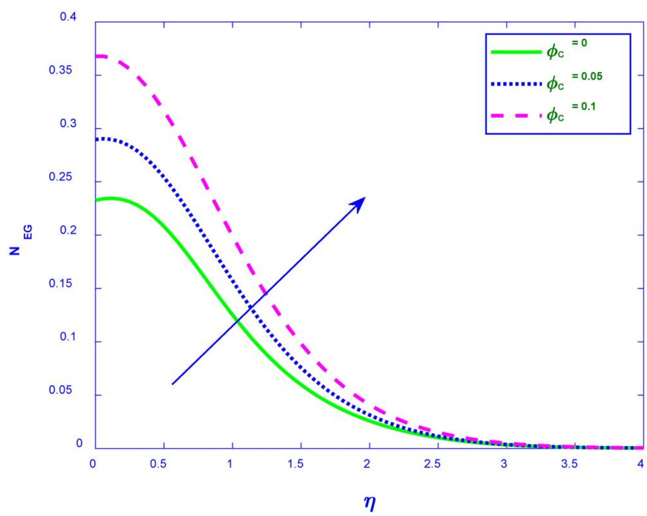

Figure 12. Situation where ϕ_C influences. N_{EG}

entropy generation is owing to fluid friction rather than heat transfer (see Fig. 13). Therefore, by increasing the nanoparticle volume fraction, engineers can design systems, where minimizing frictional losses and maximizing heat transfer are prioritized. This could be useful in applications like microfluidic devices, heat sinks, and other systems where controlling entropy generation and optimizing heat transfer characteristics are crucial for efficiency and performance. Fig. 14 illustrates a drop in the Bejan number profile with the surge in Br. More mechanical energy is transformed into thermal energy as Br increases as a result of the increased friction between the particles in the fluid. This heat generation creates thermal gradients and randomness within the system, contributing to entropy.

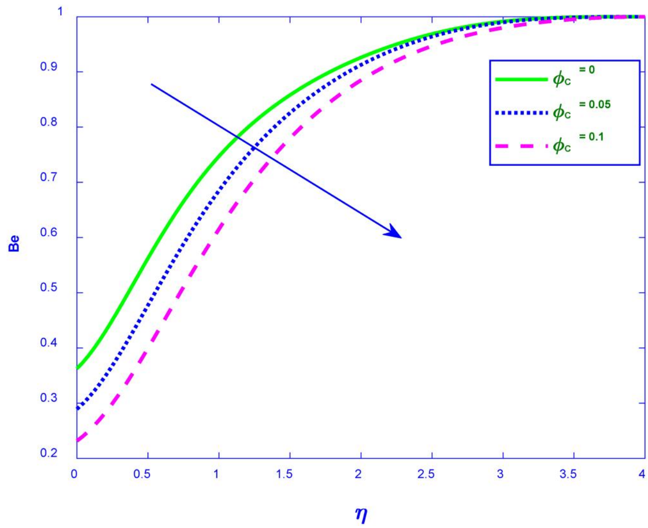

Figure 13. Situation where ϕ_C influences *Be*.

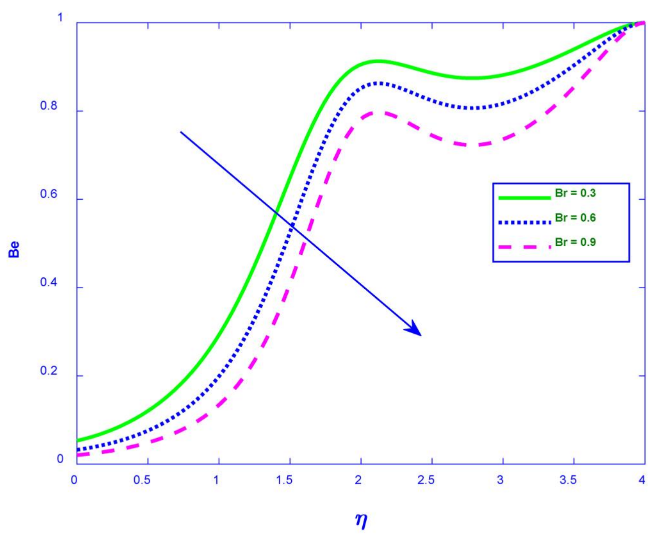

Figure 14. Situation where *Br* influences *Be*.

4.3. Multiple linear regression

Important technical characteristics, such as the mass transmission rate, and specific variables, such as the Soret number, were linked in this study using the models outlined below:

$$Cf = p_0 + p_1 We + p_2 Cs \tag{17}$$

$$Nu = q_0 + q_1 Du + q_2 \phi_C \tag{18}$$

$$Sh = r_0 + r_1 Sr + r_2 \mathrm{Kr} \tag{19}$$

We used 20 different sets of variables for each computation to speed up the process of discovering results:

$$Cf = -3.0909 + 2.476We + 0.2971Cs \tag{20}$$

$$Nu = 0.463 - 0.3128Du + 0.6438\phi_C \tag{21}$$

$$Sh = 0.2845 - 0.0141Sr + 0.3321Kr \tag{22}$$

According to Equation (20), the friction coefficient upsurges as Cs and We surges (see Fig. 15). Couple stress enhances momentum transfer within the fluid. This results in a steeper velocity gradient near the wall, contributing to higher shear stress and consequently, a higher skin friction coefficient. This result has significant implications for the aerospace and automotive industries, where skin friction drag is a major concern for fuel efficiency. By understanding the relationship between couple stress and skin friction coefficient, engineers can design more streamlined vehicles that reduce drag and improve fuel economy. It is detected that the friction factor increases by 18.34% as the couple stress parameter value falls between 0.5 and 3. According to Equation (21), the Nusselt number upsurges as ϕ_G surges and declines as Dufour number surges (see Fig. 16). When the Dufour number surges, the contribution of mass diffusion to the heat transmission becomes more substantial. This additional heat source can lead to a diminution in the temperature gradient near the surface, reducing the rate of heat transmission. In practical applications, the relationship between the Dufour number and the Nusselt number can be utilized in various engineering processes. For instance, in the design of heat exchangers, understanding this relationship can aid in optimizing the efficiency of heat and mass transfer processes. Additionally, in chemical separation techniques, manipulating the Dufour number can aid in controlling the concentration of desired components within a mixture. The Nusselt number drops by 13.30% when the Dufour number values fall between 0 and 1. According to equation [22], the Sherwood number upsurges as Kr surges and declines as Soret number surges (see Fig. 17). In fluid flow, a surge in the Soret number indicates a firmer thermal diffusion effect. This

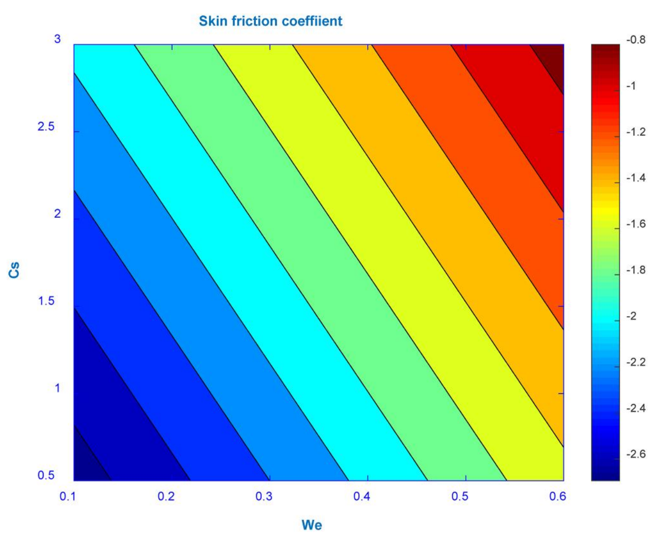

Figure 15. Situation where We and Cs influences Cf.

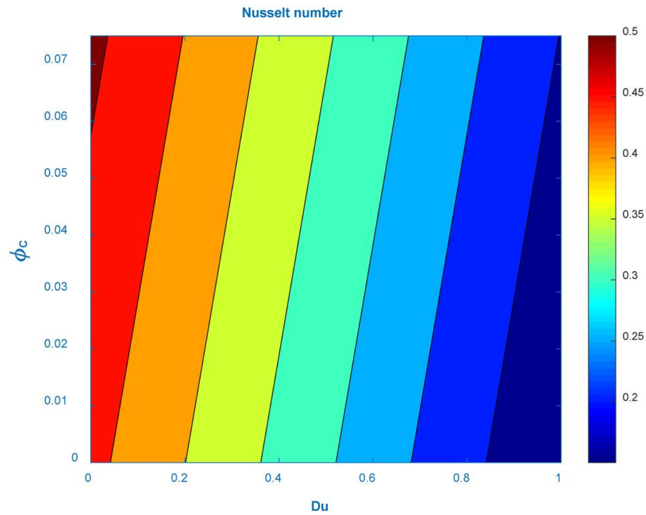

Figure 16. Situation where *Du* and ϕ_C influences *Nu*.

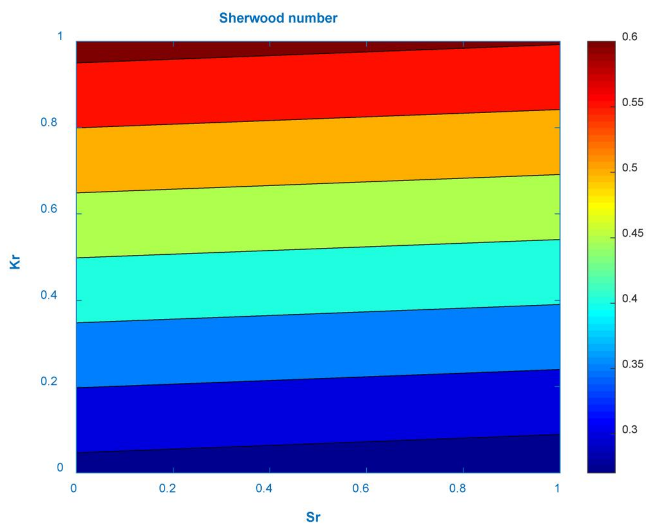

Figure 17. Situation where *Sr* and *Kr* influences *Sh*.

causes a surge in the concentration of the solute near the cooler regions of the fluid. Therefore, the concentration gradient driving the mass transfer process decreases, leading to a lower Sherwood number, which represents the mass transmission rate.

This finding has significant implications for various engineering applications, particularly in areas like chemical processing and separation technologies. For example, in processes like distillation or membrane separation, increasing the Soret number (by carefully controlling temperature gradients) could be used to decrease the Sherwood number, effectively reducing the rate of mass transfer and potentially improving the efficiency of separation. This understanding could lead to the development of more energy-efficient and precise separation techniques in various industries. The rate of mass transmission drops by 9.69% when the Soret number values fall between 0 and 1.

Table 2. Validation with former results for $-\theta'(0)$

	$-\theta'(0)$	
Pr	Ali et al. [51]	Present outcomes
1	1.0000	0.995871
2	1.5227	1.522607
3	1.9236	1.923629
4	2.2607	2.260790
5	2.5575	2.557680

5. Validation

We carry out a comparison with the past research obtained under comparable conditions to ensure the validity of our results. The results in Table 2 demonstrate that there is a high level of agreement.

6. Conclusions

An investigation is conducted into the steady movement of the CY hybrid nanofluid. The analysis examines fluid flow across an irregularly extended sheet influenced by several factors including non-Fourier heat flux, and Dufour number. Key findings are presented below:

- The study demonstrates a significant increase in the skin friction coefficient (by 18.34%) with increasing couple stress parameter.
- The Nusselt number decreases by 13.30% with increasing Dufour number.
- The Sherwood number decreases by 9.69% with increasing Soret number.
- An increase in the Weissenberg number (CY fluid parameter) leads to a reduction in fluid velocity due to enhanced shear-thinning behavior.
- An increase in the thermal relaxation time leads to a decline in fluid temperature due to slower thermal energy distribution.
- An upsurge in the Dufour number leads to a surge in entropy generation due to larger temperature gradients.
- An intensification in the nanoparticle volume fraction causes an escalation in the entropy generation due to increased fluid viscosity and frictional forces.

7. Limitations

This study, however thorough, is constrained by certain simplifying assumptions that delineate the extent of its conclusions. A primary restriction is the oversight of viscous dissipation and induced magnetic field effects, which may be substantial in high-shear flows or strong magnetic fields, potentially modifying the heat transfer and fluid dynamics characteristics. The model presumes certain spherical nanoparticles (Cu and $TiO2$) and neglects the effects of nanoparticle morphology (e.g., rods and platelets), size distribution, or aggregation, all of which significantly influence thermal conductivity and rheological properties. The analysis is restricted to a singular geometry (a nonlinear stretching sheet) and a particular fluid mixture (Water-Ethylene Glycol), thereby constraining the direct applicability of the findings to more intricate geometries such as porous cylinders, wedges, or alternative base fluids like oils or liquid metals.

8. Future research

- Investigating the influence of different nanoparticle shapes and sizes on the flow and heat transfer characteristics would be valuable.
- Extending the analysis to other geometries, such as cylinders, spheres, and wedges, to broaden the applicability of the findings.
- Exploring the behavior of other types of nanofluids, such as those with different base fluids (e.g., water-based and oil-based) or different types of nanoparticles (e.g., graphene and carbon nanotubes).
- Employing more advanced numerical techniques, such as spectral methods or lattice Boltzmann methods, to improve the accuracy and efficiency of the simulations.

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Nomenclature

σ	Electrical conductivity
Pr	Prandtl number
ϕ_C	Volume fraction of Cu
We	Weissenberg number
ρ	Density
α	Temperature difference parameter
St	Schmidt number
C_P	Specific heat
h_f	Convection heat transfer coefficient
Mg	Magnetic field Parameter
Cs	Couple stress parameter
γ_0	Couple stress coefficient
δ, Υ	Thermal relaxation parameters
Γ, d	Fluid constants
k_T	Thermal mean temperature
Kr	Chemical reaction parameter
Du	Dufour number
η	Similarity variable
ϕ_T	Volume fraction of TiO_2
σ^*	Stefan-Boltzmann constant
Br	Brinkman number
Re_x	Reynolds number
Bi	Biot number
Λ	Diffusion parameter
β	Concentration difference parameter
D_m	Molecular diffusivity
q_w	Wall heat flux
Ra	Thermal radiation parameter
m	Velocity index parameter
k^*	Mean absorption coefficient
c_s	Concentration susceptibility
T_m	Fluid mean temperature
s_w	Wall mass flux
Sr	Soret number